

4. The definition of discrete convolution is

$$z[n] \equiv (x * y)[n] \equiv \sum_{k=-\infty}^{\infty} x[k]y[n-k], \quad n \in \mathbb{Z}.$$

Prove that for the practical case, $x[0], x[1], \dots, x[p-1]$ and $y[0], y[1], \dots, y[q-1]$, this can be rewritten as

$$z[n] = \sum_{k=\max\{0, n-(q-1)\}}^{\min\{p-1, n\}} x[k]y[n-k], \quad n = 0, 1, \dots, p+q-1.$$

(a) Show that the number of multiplications for calculating the discrete convolution between two sequences of p respectively q elements is $p \cdot q$.

(b) Determine the number of additions.

7.

$$0 \leq k \leq p-1 \quad \Rightarrow \quad k \leq n$$

$$0 \leq n-k \leq q-1 \quad \Rightarrow \quad k \geq n-(q-1)$$

$$\Rightarrow k_{\min} = \max\{0, n-(q-1)\} = A$$

$$k_{\max} = \min\{n, p-1\} = B$$

$$z[n] = \sum_{k=A}^B x[k]y[n-k]$$

a) x has p elements

y has q elements

every element of x has to be multiplied by every element of $y \Rightarrow$

$$\text{nr. of multiplications} = p \cdot q$$

b) z has $p+q-1$ elements
there are $p \cdot q$ elements to be added $\} \Rightarrow$

$$\Rightarrow \text{nr. of additions} = p \cdot q - (p+q-1) = (p-1)(q-1)$$